

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - II

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Class :SYIT	Batch : 0
Date of Assignment :22/07/2026	Date/Time of Submission :22/07/2026

Questions:

2a. Write a python code to create nested tuples.

code

```
student = ("Roshan",52,("Bsc IT","Semester III"),66)
```

```
print("Name:",student[0])
```

```
print("Roll:",student[1])
```

```
print("Department:",student[2][0])
```

```
print("Semester:",student[2][1])
```

```
print("Percentage:",student[-1])
```

output

```
== RESTART: C:/Users/itla  
Name: Roshan  
Roll: 52  
Department: Bsc IT  
Semester: Semester III  
Percentage: 66  
|
```

2b. Write a python code to sort the nested tuple using sorted () function.

code

```
a = [(7, 5), (3, 8), (2, 6)]
```

```
sort = sorted(a, key=lambda x: x[0])
```

```
print(sort)
```

output

```
== RESTART: C:/Users/itlab/  
[(7, 5), (2, 6), (3, 8)]
```

```
== RESTART: C:/Users/itlab/  
[(2, 6), (3, 8), (7, 5)]
```

2c. Write a python code to copy or clone list.

code

```
list1=[1,2,3,4]
```

```
list2=list1
```

```
print("original",list2)
```

```
print("copy",list1)
```

output

```
== RESTART: C:/Users/it  
original [1, 2, 3, 4]  
copy [1, 2, 3, 4]  
|
```

2d. Write a python code to check immutability property of python tuples.

code

```
student=("roshan","nitesh","sunny","kaushal","amit")
```

```
print(student[3])
```

```
student[1]= "vijay"
```

output

```
kaushal
Traceback (most recent call last):
  File "C:/Users/itlab/Desktop/roshan syit/python for ds.py", line 4, in <module>
    student[1]= "vijay"
TypeError: 'tuple' object does not support item assignment
>|
```

Curiosity Questions:

Q1. Define a tuple. How is it different from a list? Give one example.

Q2. Write a Python program to create the following tuples:

- a) An empty tuple
- b) A tuple with integers only
- c) A tuple with mixed data types
- d) A nested tuple (tuple inside another tuple)

Q3. Write a program to access elements of a tuple using indexing and slicing.

Q4. Create a tuple and find the following:

- Length of the tuple
- Maximum and minimum value (if all are numbers)
- Count of a specific item

Q5. Write a program to convert a list into a tuple.

Q6. Write a program to sort a tuple of numbers in ascending order.

Q7. Write a program to show that tuples are immutable (cannot be changed).

Q8. Write a program to clone (copy) a list and print both original and copied lists.

Q9. Write the output of the following code:

```
t = (1, 2, 3, 4, 5)
print(t[1:4])
```

Q10. Create a tuple of student marks: (78, 85, 92, 88, 76). Write a program to:

- Find average marks
- Display marks above 80